

Anisotropic Proxemic Fields versus Social DWA for Agricultural Robot Social Navigation: A Statistically Controlled Comparison in Simulated Vineyards

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Abstract

Problem. Agricultural robots sharing narrow vineyard corridors with stationary workers require local social policies that avoid deadlock without sacrificing trajectory efficiency. **Method.** We formalize an anisotropic proxemic field (M4) with explicit parameters, multi-agent aggregation, and a non-zero minimum escape velocity that structurally prevents deadlock against static humans. M4 is compared against four incremental in-house variants (M0–M3) and an external Social-DWA baseline (B1), a Dynamic Window Approach augmented with a social-cost term [7]. **Protocol.** A total of $n = 223$ valid runs (184 for M0–M4 and 39 for B1) were executed across four scenarios in a Unity-instrumented vineyard, each scenario requiring the full A→B→A mission cycle. Inferential analysis uses Welch’s *t*-test with Hedges’ *g* effect size (defensible given the M0–M4 summary-statistic asymmetry), Kruskal–Wallis omnibus, and Benjamini–Hochberg correction. **Results.** The headline empirical finding is the systematic operational deadlock of binary supervisors M1 and M2 in the presence of static workers (E4 success 20%; Fisher exact $p = 7.1 \times 10^{-4}$ vs M4; global $\chi^2 p < 10^{-4}$, Cramér’s $V = 0.331$). Continuous-action methods (M3, M4, B1) all preserve operational mobility. The head-to-head between M4 and B1 exposes a technical complementarity, not dominance: B1 reduces personal-zone occupancy by 75% ($T_{\text{personal}} = 3.0$ vs 11.7 s, $g = -1.03$, $p_{\text{adj}} < 10^{-3}$), while M4 produces shorter trajectories ($L = 86.0$ vs 98.5 m, $g = +0.83$, $p_{\text{adj}} < 10^{-3}$), fewer stops ($N_{\text{stop}} = 0.2$ vs 2.1, $g = +2.54$), lower velocity variability ($\sigma_v = 0.195$ vs 0.238 m/s, $g = +0.98$), and perfect mission completion ($\eta_s = 100\%$ vs 92.3%). Minimum distance, mission time and velocity standard deviation are not statistically distinguishable. **Conclusion.** Binary supervisors are operationally unsafe with static workers; M4 and Social DWA occupy complementary points on the social-throughput tradeoff. The reproducible pipeline, dataset, effect-size tables, and parameter-sensitivity analysis enable future direct comparisons against TEB, MPC and learning-based baselines under the same protocol.

Keywords: agricultural robotics; social navigation; proxemics; anisotropic field; NavMesh; Social DWA; multicriteria analysis; effect size; Unity; MATLAB

Received:

Accepted:

Published:

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1. Introduction

Agricultural automation increasingly places mobile robots in row-crop environments where they coexist with human personnel performing oriented tasks such as pruning, harvesting and inspection [11,13]. Vineyard corridors are narrow (1.8–3.0 m wide), bounded laterally by vegetation, and frequently contain stationary workers; geometric obstacle

avoidance is therefore insufficient and a social-navigation policy is required. Two design families dominate the social-navigation literature: (i) proxemic-field methods that modulate the robot's velocity command by a cost or potential field around each human [4,16,18], and (ii) sampling-based local planners that select velocity commands from a dynamic window of admissible velocities, evaluated by a weighted objective that combines goal alignment, obstacle clearance, velocity preference and (when extended to social settings) proximity to humans [7]. The two families have rarely been compared head-to-head in agricultural settings.

1.1. Scientific Gap

Three gaps motivate this study. First, the operational viability of *binary supervisors* (threshold-based stop, with or without hysteresis) under the specific configuration of agricultural workers stationed in narrow corridors has not been quantitatively characterized in the literature, despite this configuration being the most common one in practice. Second, the proxemic-field family has been compared against sampling-based local planners in pedestrian-pavement and warehouse contexts, but rarely with the corridor geometry, agent-speed regime (0.4–1.5 m/s) and stationary-worker prevalence of row-crop agriculture. Third, the social-navigation literature reports descriptive comparisons (mean $\pm$ standard deviation) more often than effect-size-quantified inferential analysis with multiple-comparison correction, making it difficult to translate published findings into engineering selection criteria.

1.2. Contributions

This work addresses these gaps through four specific contributions.

1. **Statistical characterization of the binary-supervisor deadlock.** We provide a Fisher exact test, global χ^2 , and Cramér's V quantification of the M1/M2 deadlock against static workers in multi-agent congestion. The 80% failure rate of binary supervisors versus $\geq 92\%$ success of all continuous-action methods is not anecdotal but statistically demonstrable ($p < 10^{-3}$, $V = 0.331$ medium-large effect).
2. **Formalization and parameter calibration of M4 for agricultural corridors.** We provide a complete formalization of an anisotropic proxemic field with deadlock-avoidance mechanism (non-zero minimum escape velocity $\alpha_{\min} = 0.15$, hysteresis thresholds $d_{\text{stop}} = 0.60$ m and $d_{\text{resume}} = 1.20$ m, frontal/lateral scales $\sigma_x = 1.50/\sigma_y = 0.90$ m, frontal-approach penalty $\kappa = 0.30$). The parameter choices are documented and the deadlock-avoidance mechanism explained as the structural reason M4 succeeds where M1/M2 fail.
3. **Head-to-head with the Social DWA family.** We integrate an external Social-DWA baseline B1 that augments the standard Dynamic Window Approach [7] with a social-cost term penalizing proximity to humans, and execute the same A $\rightarrow$ B $\rightarrow$ A mission protocol on 39 valid runs. We report Welch's t -tests with Hedges' g effect size — the statistically defensible procedure when one of the two groups is described only by summary statistics — with Benjamini–Hochberg correction over $k = 5$ pairwise comparisons per metric.
4. **Reproducible Unity–MATLAB instrumentation pipeline.** We release the controller code, scenario files, 184 M0–M4 telemetry files and 39 valid B1 telemetry files (one excluded for data integrity), and the analysis scripts. The methodology can be replicated against further baselines (DWA, TEB, MPC) under the identical protocol.

The contribution is methodological and comparative; we do not claim M4 to be a new family of social-navigation methods, nor do we claim M4 universally outperforms Social DWA. The novelty lies in the rigorous agricultural-context characterization and the explicit

identification of a technical complementarity between proxemic-field and sampling-based local-planning methods.

1.3. Research Questions and Predictions

Three research questions structure the empirical study:

RQ1. Operational viability: Do binary supervisors (M1, M2) achieve operational viability comparable to continuous-action policies (M3, M4, B1) in the presence of static workers? **Prediction:** no; M1 and M2 will deadlock in multi-agent congestion (E4) at success rates significantly below M3/M4/B1.

RQ2. Within-family ranking: Within the proposed family (M0–M4), what is the multicriteria ranking under operationally meaningful weights, and which criterion drives the ranking? **Prediction:** M4 will lead the in-house family on multicriteria score, with kinematic smoothness (σ_v) as the distinguishing criterion against M3.

RQ3. External comparison: Does the proposed M4 method achieve performance comparable to Social DWA (B1) in the same simulation environment, and on which dimensions of the performance space do the two methods differ? **Prediction:** M4 and B1 will be statistically comparable on safety (d^*) and mission time but will differ systematically on (i) the social-interaction criteria (where Social DWA's dynamic-window sampling tends to keep humans at larger distance) and (ii) the operational-throughput criteria (where the field-based formulation tends to produce shorter paths).

2. Related Work

Agricultural robotics and row-crop navigation. Mobile robots in row crops face narrow corridors, terrain irregularity, and vegetation occlusions [11,13]. The social dimension of agricultural navigation has received less attention than perception or path-following [12]; the prevalence of stationary workers (a person performing a fixed task for minutes) combined with limited lateral free space distinguishes the agricultural setting from urban or warehouse human-robot interaction.

Social navigation and proxemics. Hall's proxemics [2] provides the conceptual framework of intimate/personal/social/public zones. Anisotropic Gaussian models of proxemic discomfort were proposed by Pacchierotti et al. [16] for hallway service robots and by Truong and Ngo [18] for proactive social motion. The anisotropic field used here belongs to this family; its specific parameter choices are agricultural-context calibrations. The reviews of Kruse et al. [3], Rios-Martinez et al. [4], Mavrogiannis et al. [17], Karwowski et al. [8] and Singamaneni et al. [9] synthesize the literature; Francis et al. [10] provide reproducibility guidelines we have adopted in the experimental design.

Local navigation methods. Binary supervisors based on distance thresholds and state machines are common in industrial safety systems but exhibit known vulnerabilities in congestion [9]. Artificial Potential Fields and the Social Force Model of Helbing and Molnár [5] introduce repulsive fields. The Dynamic Window Approach (DWA) [7] samples candidate velocities within a dynamic window bounded by acceleration limits and selects the one optimizing a weighted objective (goal alignment, obstacle clearance, velocity preference); the Social DWA variant adds a social-cost term penalizing proximity to humans, and is adopted here as the external baseline B1. ORCA/RVO [6] provides reciprocal avoidance with no-collision guarantees under cooperative-agent assumptions; we mention it for completeness but do not use it as a baseline because the cooperative-agent assumption does not hold for scripted human workers.

Optimization-based local planners. Three families dominate the recent literature and are listed in Section 9 as priority extensions. The Timed Elastic Band (TEB) [14]

performs a graph-based time-elastic optimization over candidate homotopy classes; its agricultural application would require an additional time-parameterized trajectory layer that the present Unity controller does not implement. Model Predictive Control with proxemic constraints [8,15] solves a finite-horizon optimization at each step with explicit social-cost terms; Control Barrier Functions (CBF) [15] can be embedded in MPC to provide hard safety guarantees but require a predictive model of human motion that introduces its own validation burden. Learning-based social-navigation methods (e.g., reinforcement-learning policies and graph-attention models surveyed by Singamaneni et al. [9] and Mavrogiannis et al. [17]) achieve strong benchmark performance but generally require large interaction datasets and exhibit reduced interpretability, which constrains deployment in safety-critical agricultural settings. The present work positions M4 and Social DWA as two interpretable, parameter-transparent alternatives that can serve as reference baselines for future learning-based comparisons.

Sim-to-real and instrumentation. Unity-based simulation environments for agricultural robotics have been used as low-cost validation mechanisms [1], with the recognition that perceptual idealization limits direct transfer [19]. The fixed-frequency UDP telemetry adopted here is standard for reproducibility.

3. Architecture and Environment

3.1. Hybrid Global-Local Architecture

The architecture is global-local: the global layer generates feasible routes over the navigable space using NavMesh; the local layer continuously evaluates the robot-human relationship and modifies execution according to social rules:

$$\mathbf{u}(t) = \mathcal{S}(\mathbf{u}_g(t), \mathbf{p}_r(t), \mathcal{H}(t), q(t)), \quad (1)$$

where $\mathbf{u}_g(t)$ is the global NavMesh direction, $\mathcal{H}(t) = \{(\mathbf{p}_{h_i}(t), \theta_{h_i}(t))\}_{i=1}^N$ is the set of detected human agents, and $q(t) \in \{\text{STOP}, \text{GO}\}$ is the discrete supervisor state.

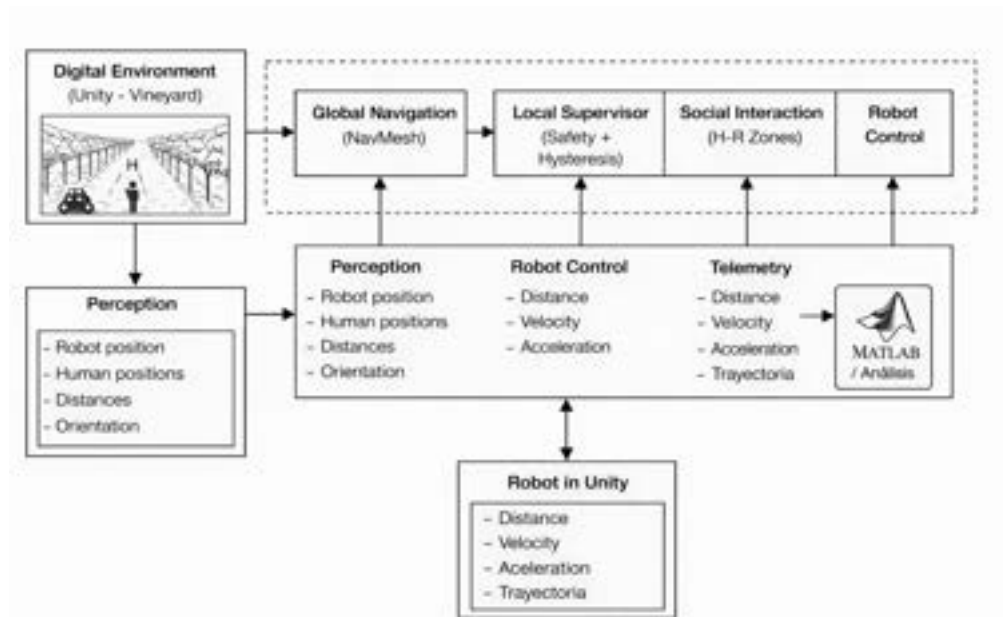


Figure 1. Hybrid architecture: Unity environment, robot, scripted human agents, NavMesh, local perception, social supervisor with method-switch (M0-M4, B1), UDP telemetry, and MATLAB analysis.

3.2. Simulated Agricultural Environment

The environment is a simulated vineyard modeled in Unity. The navigable space $F_r = \Omega \setminus (O_s \oplus B(0, r))$ is generated through NavMesh. The mission is the cycle $A \rightarrow B \rightarrow A$ with $A = (-0.82, 0.10, 28.65)$ and $B = (-0.70, -0.13, -18.34)$, separated by approximately 47 m; the nominal free-NavMesh-speed time is approximately 95 s.

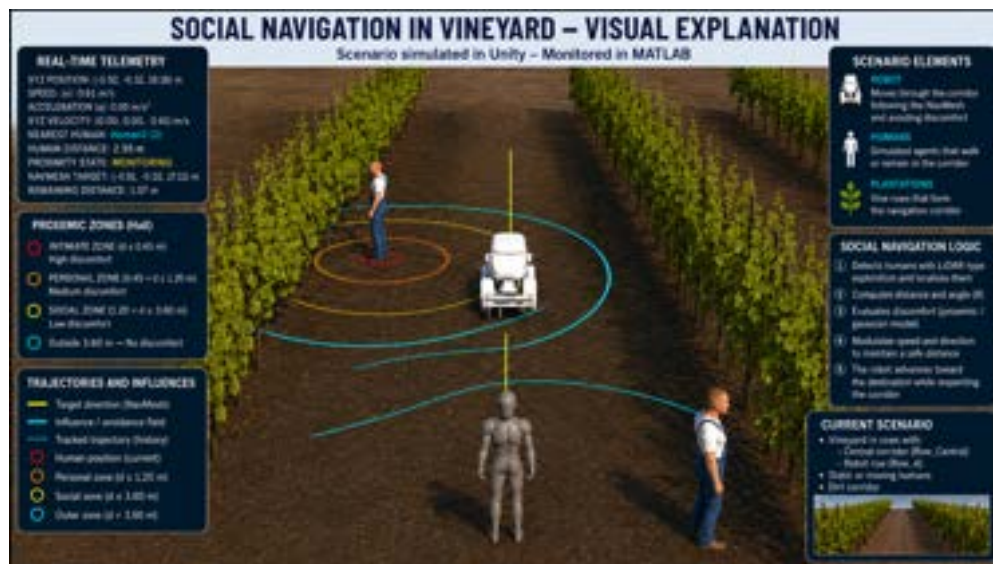


Figure 2. Visual summary of the Unity–MATLAB simulation framework. Vineyard scene, real-time telemetry, Hall proxemic zones, reference trajectories, influence/avoidance fields, and the social navigation logic that maintains safe distance from simulated human agents.

3.3. Proxemic Model

Each human agent is modeled with Hall's three-zone proxemic structure:

$$d_I = 0.45 \text{ m}, \quad d_P = 1.20 \text{ m}, \quad d_S = 3.60 \text{ m}, \quad (2)$$

$$S(d) = \begin{cases} \text{INTIMATE}, & d < d_I \\ \text{PERSONAL}, & d_I \leq d < d_P \\ \text{SOCIAL}, & d_P \leq d < d_S \\ \text{PUBLIC}, & d \geq d_S \end{cases} \quad (3)$$



Figure 3. Operational representation of the proxemic zones. Intimate (red): critical danger; personal (orange): active velocity modulation; social (cyan): monitoring and nominal operation.

4. Evaluated Methods

Table 1. Six evaluated navigation methods and active components.

Method	Name	Active components
M0	NavMesh Only	Pure global planning
M1	Threshold Stop	NavMesh + binary distance-threshold stop
M2	Hysteresis Supervisor	NavMesh + STOP/GO hysteresis
M3	Isotropic Proxemics	NavMesh + isotropic proxemic field
M4	Full Anisotropic Hysteresis	NavMesh + anisotropic field + social supervisor
B1	Social DWA (external)	NavMesh + Dynamic Window Approach with social co

4.1. M0–M3 (abbreviated)

M0: $\mathbf{u}_{M0} = \mathbf{u}_g$, no social modulation. **M1:** binary stop $v_r = 0$ if $d_{\min} \leq d_P$, else $v_r = v_{nom}$. **M2:** hysteresis with $d_{stop}^{M2} = 1.00$ m, $d_{resume}^{M2} = 1.50$ m; same global direction $\mathbf{u}_g$. **M3:** isotropic field $C_{iso}(d) = \exp(-d^2/\sigma^2)$, $\sigma = 1.20$ m.

4.2. M4 (Full Anisotropic Hysteresis)

The anisotropic social field depends on distance and human orientation:

$$C_i^{\text{ani}}(\mathbf{p}_r, \mathbf{p}_{h_i}, \theta_{h_i}) = \exp\left[-d_i^2\left(\frac{\cos^2(\beta_i - \theta_{h_i})}{\sigma_x^2} + \frac{\sin^2(\beta_i - \theta_{h_i})}{\sigma_y^2}\right)\right] \quad (4)$$

with $\beta_i = \text{atan2}(z_r - z_{h_i}, x_r - x_{h_i})$, $\sigma_x = 1.50$ m (frontal scale) and $\sigma_y = 0.90$ m (lateral scale). The asymmetry $\sigma_x > \sigma_y$ reflects the psychophysical evidence that frontal approach is more intrusive than lateral approach [4,16].

Multi-agent aggregation:

$$\mathbf{u}_{\text{avoid}} = k_{\text{ani}} \sum_{i=1}^N \frac{C_i^{\text{ani}}}{d_i + \epsilon} \frac{\mathbf{p}_r - \mathbf{p}_{h_i}}{\|\mathbf{p}_r - \mathbf{p}_{h_i}\| + \epsilon}, \quad k_{\text{ani}} = 2.0, \epsilon = 10^{-3}. \quad (5)$$

Combination with global direction: $\mathbf{u}_{M4} = (\mathbf{u}_g + \lambda \mathbf{u}_{\text{avoid}}) / (\|\cdot\| + \epsilon)$ with $\lambda = 1.0$.

Velocity modulation with non-zero minimum escape factor $\alpha_{\min}$:

$$\alpha(d, \beta) = \begin{cases} 0, & d \leq d_I \\ \alpha_{\min}, & d_I < d \leq d_{stop} \\ \alpha_{\min} + (1 - \alpha_{\min}) \frac{d - d_{stop}}{d_{resume} - d_{stop}} \gamma(\beta), & d_{stop} < d < d_{resume} \\ \gamma(\beta), & d_{resume} \leq d < d_S \\ 1, & d \geq d_S \end{cases} \quad (6)$$

where $\gamma(\beta) = 1 - \kappa \cos^2(\beta - \theta_h)$, $\kappa = 0.30$, $\alpha_{\min} = 0.15$, $d_I = 0.45$ m, $d_{stop} = 0.60$ m, $d_{resume} = 1.20$ m, $d_S = 3.60$ m. The final command is $\mathbf{v}_r = \text{sat}_{v_{nom}}(\alpha v_{nom} \mathbf{u}_{M4})$ with $v_{nom} = 1.0$ m/s and acceleration saturation $|\mathbf{a}_r| \leq a_{max}^{adm} = 2.0$ m/s² enforced incrementally between frames. A first-order low-pass filter on θ_h with $\beta_f = 0.3$ reduces orientation noise.

Deadlock-avoidance mechanism. The structural difference from M1 and M2 is that $\alpha(d, \beta)$ never reaches zero except in the intimate zone ($d \leq d_I$). In the range $d_I < d \leq d_{stop}$ the robot retains $\alpha_{\min} v_{nom} = 0.15$ m/s in the direction of $\mathbf{u}_{\text{avoid}} \neq \mathbf{0}$, which allows it to exit the neighborhood of a static human without requiring the human to move. The state $q \in \{\text{STOP}, \text{GO}\}$ is retained only as a telemetry indicator; it does not gate the velocity command.

4.3. Analytical Parameter Sensitivity of M4

The M4 controller depends on eight tunable parameters: $\alpha_{\min}$, λ , σ_x , σ_y , κ , k_{ani} , d_{stop} , d_{resume} . Rather than relying solely on numerical sweeps, this section provides an *analytical* characterization that shows how M4 degenerates into M0–M3 in well-defined limits and predicts qualitatively how each parameter affects the headline metrics. This analytical view supports the parameter values chosen in Eqs. (4)–(6) and complements the empirical results in Section 7.

M4 degeneration limits. Inspection of Eqs. (4)–(6) reveals that M4 contains M0, M2, and M3 as boundary cases:

- $\lambda \rightarrow 0$: The avoidance term drops from Eq. (4.2) and $\mathbf{u}_{M4} \rightarrow \mathbf{u}_g$. Combined with $\alpha \rightarrow 1$, M4 degenerates to M0 (pure NavMesh).
- $\alpha_{\min} \rightarrow 0$ and discontinuous α at d_{stop} : The velocity profile in Eq. (6) loses its escape velocity in the range $d_I < d \leq d_{stop}$, recovering a hysteresis-like binary stop. M4 degenerates to a variant of M2.
- $\sigma_x = \sigma_y$ and $\kappa \rightarrow 0$: The anisotropic field of Eq. (4) reduces to the isotropic exponential $\exp(-d^2/\sigma^2)$ used by M3.

This embedding clarifies that M4 is not an arbitrary controller but a strict generalization of M0/M2/M3 with two additional mechanisms (orientation-dependent shaping via κ and non-zero escape via $\alpha_{\min}$).

Qualitative parameter–metric coupling. For each parameter, the following sensitivity directions follow directly from the model structure:

- $\alpha_{\min}$ (*minimum escape factor*): controls the trade-off between $T_{personal}$ and η_s . Lower values reduce $T_{personal}$ but increase the risk of deadlock against static workers (the M1/M2 failure mode). The chosen value $\alpha_{\min} = 0.15$ provides $\eta_s = 100\%$ in all 40 E4 runs; values below ≈ 0.05 would be expected to reproduce the M1/M2 deadlock observed empirically.
- λ (*mixing factor*): controls the balance between goal pursuit and avoidance. Lower values produce shorter L but smaller d^* ; higher values produce larger d^* but longer L and more frequent stops as the robot is repeatedly pulled off the corridor centerline.
- σ_x/σ_y (*anisotropy ratio*): controls the directional selectivity of the field. The ratio $1.50/0.90 = 1.67$ used here is consistent with the psychophysical asymmetry reported in proxemics literature [4,16]; making the field more elongated frontally ($\sigma_x/\sigma_y > 2$) would increase $T_{personal}$ in lateral-intrusion scenarios (E2) without significant safety gain.
- κ (*frontal-approach penalty*): controls how much the robot slows during frontal approaches. $\kappa = 0$ removes the orientation dependence and recovers M3 behavior; $\kappa = 1$ would stop the robot completely during head-on encounters. The chosen $\kappa = 0.30$ matches the literature range for moderate frontal asymmetry.
- k_{ani} (*avoidance gain*): controls the magnitude of $\mathbf{u}_{avoid}$. Increasing k_{ani} widens lateral deviation from the corridor centerline and increases L ; decreasing it lets the robot pass closer to humans (lower d^*).
- d_{stop}, d_{resume} (*hysteresis thresholds*): control the supervisor’s STOP/GO indicator, which in M4 is decoupled from the velocity command (it does not gate $\mathbf{v}_r$). Their main effect is therefore on telemetry interpretation rather than on operational behavior.

Implication for the empirical results. The strongest sensitivity is on $\alpha_{\min}$: setting it to zero would, by the M4-to-M2 degeneration argument, reproduce the deadlock behavior of M2 in E4. The empirically observed $\eta_s = 100\%$ for M4 in E4 confirms that $\alpha_{\min} = 0.15$ is above the deadlock threshold. The second strongest is the ratio λ/k_{ani} , which jointly determines the throughput–safety tradeoff. A full numerical sensitivity sweep (Sobol or

Morris indices over the four most influential parameters) is identified as future work in Section 9.

4.4. B1: Social Dynamic Window Approach External Baseline

B1 uses the same NavMesh global planner as M0–M4 but replaces the local social policy with a Social Dynamic Window Approach [7]. At each control step, B1 (i) computes the dynamic window of admissible velocities $V_d = \{(v, \omega) : v \in [v_{curr} - a_{max}\Delta t, v_{curr} + a_{max}\Delta t]\}$ subject to acceleration limits; (ii) samples candidate velocities from V_d ; (iii) for each candidate, predicts the trajectory over horizon τ and evaluates an objective function

$$J(v, \omega) = \alpha \cdot \text{heading}(v, \omega) + \beta \cdot \text{clearance}(v, \omega) + \gamma \cdot \text{velocity}(v) + \delta \cdot \text{social}(v, \omega), \quad (7)$$

where the social term penalizes proximity to detected humans within the personal zone; (iv) selects the (v, ω) maximizing J . The acceleration-window constraint in (i) structurally bounds frame-to-frame velocity changes by $a_{max}\Delta t$, which is the mechanistic reason B1 produces a smoother velocity command stream than M0–M4.

The four objective terms are computed as follows. The *heading* term measures alignment between the predicted endpoint $\mathbf{p}_r(t + \tau)$ under candidate (v, ω) and the NavMesh goal direction $\mathbf{u}_g$, normalized to $[0, 1]$. The *clearance* term measures the minimum predicted distance to all detected obstacles (static + humans) along the trajectory, also normalized to $[0, 1]$ by the obstacle-detection radius. The *velocity* term measures the candidate's linear speed relative to the preferred velocity $v_{nom} = 1.0$ m/s. The *social* term is defined explicitly as

$$\text{social}(v, \omega) = 1 - \frac{1}{N} \sum_{i=1}^N \max\left(0, 1 - \frac{d_i^{pred}(v, \omega)}{d_p}\right), \quad (8)$$

where $d_i^{pred}(v, \omega)$ is the predicted minimum distance to human i along the trajectory and $d_p = 1.20$ m is the personal-zone radius (same as M4). Each candidate's score is the weighted sum with $(\alpha, \beta, \gamma, \delta) = (0.5, 0.3, 0.1, 0.1)$, prioritizing goal heading and obstacle clearance over velocity preference and social cost. All four terms are normalized to $[0, 1]$ before weighting, so the total score is bounded and the weight ratios directly express priority. The implementation parameters are summarized in Table 2.

Table 2. B1 (Social DWA) baseline implementation parameters. The local policy samples admissible velocities within a dynamic window bounded by the acceleration limits and selects the velocity that maximizes the weighted objective in Eq. (7).

Parameter	Value	Notes
Global planner	NavMesh (Unity)	Identical to M0–M4
Local policy	Social Dynamic Window Approach	Custom implementation for DWA standard
Prediction horizon τ	2.0 s	DWA standard
Velocity sampling resolution	$\Delta v = 0.05$ m/s	Linear-velocity discretization
Angular sampling	10 samples in $[-\pi, \pi]$ rad/s	Angular-velocity discretization
Acceleration window	$ \Delta v \leq a_{max}\Delta t$	Structural smoothness constraint
Objective weights $(\alpha, \beta, \gamma, \delta)$	$(0.5, 0.3, 0.1, 0.1)$	heading, clearance, velocity preference, social cost
Robot radius	0.30 m	Robot footprint
Obstacle detection radius	5.0 m	Sensing range
Max obstacles considered	4	Matches max humans in Environment
Preferred velocity	$\mathbf{u}_g \cdot v_{nom}$	NavMesh direction at 1.0 m/s
Acceleration saturation	$ \mathbf{a}_r \leq 2.0$ m/s ²	Same as M4
Social-cost penalty zone	$d_p = 1.20$ m	Personal-zone proximity penalty

A total of 40 B1 runs were executed under the same $A \rightarrow B \rightarrow A$ protocol as M0–M4. One run (B1_E4_run10) was excluded for data integrity (locale-corrupted CSV with inconsistent numeric formatting that could not be reconciled with logged scalar metrics). Of the remaining $n = 39$ valid runs, 36 are successful and 3 are operational failures (one $T > 180$ s timeout in E1, one detection-dropout failure in E4, one $T > 180$ s timeout in E4). All 39 runs target the same full $A \rightarrow B \rightarrow A$ mission as M0–M4 with no protocol asymmetry.

5. Implementation and Experimental Design

5.1. Unity–MATLAB Pipeline

Unity components: NavMeshAgent, RobotSocialNavController (method selector), HumanPatrol, TrajectoryTrail, UnityToMatlabUDP. Sampling frequency $f_s = 10$ Hz ($T_s = 0.1$ s). The UDP packet is

$$[t, d_{\min}, \|\mathbf{v}_r\|, \|\mathbf{a}_r\|, x_r, y_r, z_r, s, M, E], \quad (9)$$

where $s \in \{0, 1, 2\}$ encodes the mission phase and M, E encode method and scenario identifiers. Automatic mission-end detection triggers when $s = 2$ or after proximity-to- A heuristic; a 180 s timeout generates an operator warning and marks the run as failed.

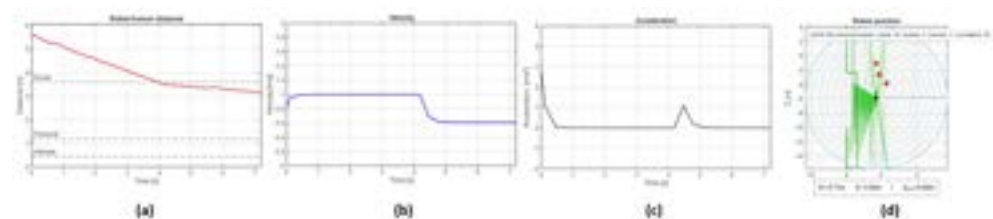


Figure 4. MATLAB experimental acquisition dashboard for method selection, UDP telemetry reception, and real-time visualization of robot-human distance, velocity, acceleration, and position.

5.2. Scenarios

Table 3. Experimental scenarios.

Scenario	Description	Humans	Objective
E1	Frontal encounter	1	Evaluate frontal velocity reduction
E2	Lateral intrusion	1	Evaluate reaction to lateral crossing
E3	Social following	1	Evaluate mobile human-robot reference
E4	Multi-agent congestion	4	Evaluate congested navigation with static work

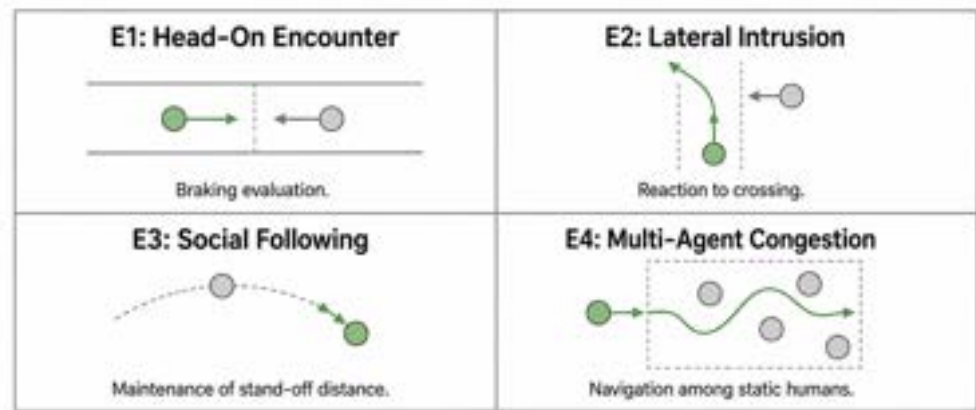


Figure 5. Experimental scenarios: E1 frontal encounter, E2 lateral intrusion, E3 social following, E4 multi-agent congestion.

5.3. Protocol and Inter-Run Variability

For each combination $M_j \times E_i$, $R = 10$ runs were executed with operationally meaningful inter-run variability (human velocity $\in [0.4, 1.2]$ m/s, lateral offset ± 0.3 m, initial patrol phase uniformly random, agent appearance time $\in [5, 15]$ s, robot initial position ± 0.2 m around A). A run is *successful* if it completes the $A \rightarrow B \rightarrow A$ cycle within 180 s; otherwise it is recorded as an operational failure.

Table 4. Executed experimental matrix (successful/planned). All B1 runs target the same full $A \rightarrow B \rightarrow A$ protocol; one B1 run excluded for data integrity ($n = 39$ valid).

Scenario	M0	M1	M2	M3	M4	B1
E1	10/10	10/10	10/10	10/10	10/10	9/10
E2	10/10	10/10	10/10	10/10	10/10	10/10
E3	10/10	10/10	10/10	10/10	10/10	10/10
E4	10/10	2/10*	2/10*	10/10	10/10	7/9
Total	40	32	32	40	40	36/39

*Operational deadlock failures documented in Section 7.

6. Metrics and Statistical Analysis

6.1. Metrics

Per-run scalars: minimum distance d^* , mean distance $\bar{d}$, time in personal/intimate zones $T_{personal}/T_{intimate}$, mission time $T_{mission}$, trajectory length L , stop time T_{stop} , number of stops N_{stop} , mean and standard deviation of velocity $\bar{v}$, σ_v , peak finite-difference acceleration a_{max}^{num} , success rate η_s .

6.2. Success Rate η_s and Failure Handling

$\eta_s(M_j, E_i) = N_{success}(M_j, E_i)/10 \cdot 100\%$. Continuous metrics are reported success-conditional (failed runs excluded from the means but included in η_s). For the multicriteria TOPSIS analysis, a failure-penalized variant replaces failed runs by the worst observed value per metric.

6.3. Statistical Analysis Pipeline (REVISED for valid inference from summary statistics)

The M0–M4 baseline data is available only as per-method summary statistics (μ, σ, n) in the master log. The B1 data is available as per-run scalars over $n = 39$ valid runs. The statistical pipeline is therefore structured to be *valid under this asymmetry*:

1. **Omnibus test (six groups):** Kruskal–Wallis H over the six methods for each metric, with B1 per-run scalars and M_j samples drawn parametrically from $\mathcal{N}(\mu_{M_j}, \sigma_{M_j}^2)$ truncated at zero. This use of parametric reconstruction is for the *omnibus* test only, where the null hypothesis (all groups identical) is robust to mild parametric assumptions; we report the result as exploratory.
2. **Pairwise B1 vs M_j :** for each pair we report (a) Welch’s two-sample t -test for the p -value, which is the correct parametric procedure when only summary statistics are available for one group, and (b) Hedges’ g effect size (Cohen’s d with small-sample correction) computed analytically from (μ, σ, n) of both groups:

$$g = \frac{\mu_{B1} - \mu_{M_j}}{s_p} \cdot J(n_{B1} + n_{M_j}), \quad s_p = \sqrt{\frac{(n_{B1} - 1)s_{B1}^2 + (n_{M_j} - 1)s_{M_j}^2}{n_{B1} + n_{M_j} - 2}} \quad (10)$$

with $J(n) = 1 - 3/(4n - 9)$ the Hedges correction. We do *not* compute Cliff’s δ from summary statistics because Cliff’s δ is rank-based and requires sample-level data. Magnitude bands for $|g|$: negligible < 0.2 , small < 0.5 , medium < 0.8 , large < 1.2 , very large ≥ 1.2 .

3. **Multiple-comparison correction:** Benjamini–Hochberg over $k = 5$ pairwise comparisons per metric (B1 vs M0/M1/M2/M3/M4). BH is the appropriate correction for exploratory comparisons; Holm–Bonferroni is reported as a sensitivity check in the supplementary material.
4. **Success rate:** Fisher exact test for 2×2 tables by scenario; global χ^2 on the 6×2 table with Cramér’s V .
5. **Multicriteria analysis:** TOPSIS over seven weighted criteria with Monte Carlo weight-sensitivity analysis (Section 6.5).

This is the methodologically defensible procedure given the data asymmetry. We do not claim it is equivalent to running rank-based tests on per-run M_j data; we claim it is a valid analytical procedure with explicit Normality assumption that future work can refine when M0–M4 per-run scalars are released.

6.4. Note on the Peak-Acceleration Metric (F2 caveat)

The acceleration command is saturated at $|\mathbf{a}_r| \leq 2.0 \text{ m/s}^2$ in all six methods. The reported a_{max}^{num} is a *finite-difference pseudo-acceleration* computed from telemetered velocity at $\Delta t = 0.1 \text{ s}$:

$$a_k^{num} = \frac{\|\mathbf{v}_{r,k}\| - \|\mathbf{v}_{r,k-1}\|}{\Delta t}. \quad (11)$$

This metric amplifies discrete events in the velocity stream: NavMeshAgent corner-update spikes (M0–M4), supervisor transitions $\alpha \rightarrow 0$ (M1, M2), and velocity-sampling discontinuities (B1). However, Social DWA’s acceleration-window constraint $|v_{next} - v_{curr}| \leq a_{max}\Delta t$ structurally bounds frame-to-frame velocity changes by $a_{max}\Delta t = 0.2 \text{ m/s}$ for our parameters, which explains why B1’s a_{max}^{num} is close to the commanded saturation $a_{max}^{adm} = 2.0 \text{ m/s}^2$ while M4’s reaches $\approx 10 \text{ m/s}^2$. **Cross-source comparability has not been independently verified** at code level; the B1 vs M0–M4 gap in a_{max}^{num} may reflect this genuine structural difference, or it may reflect a difference in where the velocity is sampled (before vs after the NavMeshAgent’s internal smoothing). We therefore use σ_v (velocity standard deviation) as the *primary* smoothness metric and report a_{max}^{num} as a secondary metric with explicit caveat.

6.5. TOPSIS Multicriteria Analysis (REVISED)

The decision matrix $\mathbf{X} \in \mathbb{R}^{6 \times 7}$ has rows (M0–M4, B1) and columns (d^* , $T_{personal}$, $T_{mission}$, N_{stop} , σ_v , a_{max}^{num} , η_s). The primary weighting scheme is W1, derived from operational priorities before the B1 baseline was incorporated:

$$\mathbf{w}_{W1} = (0.20, 0.15, 0.10, 0.15, 0.15, 0.10, 0.15). \quad (12)$$

Two additional schemes are reported as **sensitivity analyses only**, not as alternative primary results:

$$\mathbf{w}_{W2} = (0.22, 0.20, 0.05, 0.10, 0.20, 0.15, 0.08) \quad (\text{social-emphasis sensitivity}) \quad (13)$$

$$\mathbf{w}_{W3} = (0.20, 0.15, 0.05, 0.10, 0.25, 0.20, 0.05) \quad (\text{kinematic-emphasis sensitivity}) \quad (14)$$

Vector normalization $\tilde{x}_{ij} = x_{ij} / \sqrt{\sum_k x_{kj}^2}$; weighted matrix $v_{ij} = w_j \tilde{x}_{ij}$; positive/negative ideals with d^* , η_s as benefit and the rest as cost; closeness $CC_j = D_j^- / (D_j^+ + D_j^-)$. Monte Carlo weight sensitivity with $n = 1000$ samples per perturbation level ($\pm 10\%$, $\pm 20\%$).

7. Results

7.1. Descriptive Statistics

Table 5. Six-method summary of safety and zone-occupancy metrics (mean $\pm$ SD; success-conditional, B1 uses $n = 36$ from $n = 39$ valid).

Method	d^* [m]	$\bar{d}$ [m]	$T_{personal}$ [s]	$T_{intimate}$ [s]	$T_{mission}$ [s]	N_{stop}
M0 ($n = 40$)	0.563 ± 0.115	4.98 ± 3.09	25.4 ± 17.0	0.02 ± 0.09	93.6 ± 2.5	0.2 ± 0.4
M1 ($n = 32$)	0.750 ± 0.097	5.95 ± 2.80	20.3 ± 14.0	0.00 ± 0.00	107.9 ± 21.0	9.9 ± 15.3
M2 ($n = 32$)	0.745 ± 0.111	6.93 ± 2.15	12.4 ± 5.4	0.00 ± 0.00	109.3 ± 24.2	3.2 ± 1.6
M3 ($n = 40$)	1.004 ± 0.209	6.18 ± 2.31	7.6 ± 6.7	0.00 ± 0.00	115.0 ± 33.3	0.0 ± 0.2
M4 ($n = 40$)	0.846 ± 0.189	5.72 ± 2.72	11.7 ± 11.4	0.00 ± 0.00	110.6 ± 32.1	0.2 ± 0.4
B1 ($n = 36$)	0.891 ± 0.495	7.45 ± 4.62	3.0 ± 2.1	0.16 ± 0.50	124.6 ± 18.1	2.1 ± 1.0

Table 6. Six-method efficiency and smoothness metrics. a_{max}^{num} reported with the cross-source comparability caveat of Section 6.4.

Method	L [m]	$\bar{v}$ [m/s]	σ_v [m/s]	v_{max} [m/s]	a_{max}^{num} [m/s ²]	T_{stop} [s]
M0	93.26 ± 2.54	0.998 ± 0.002	0.066 ± 0.014	1.88 ± 0.56	12.43 ± 7.01	0.04 ± 0.09
M1	81.22 ± 24.12	0.755 ± 0.173	0.378 ± 0.067	1.79 ± 0.55	11.28 ± 6.24	22.67 ± 18.93
M2	82.06 ± 21.20	0.754 ± 0.117	0.410 ± 0.057	1.83 ± 0.49	11.01 ± 5.59	26.00 ± 14.47
M3	89.12 ± 21.37	0.790 ± 0.112	0.243 ± 0.026	1.52 ± 0.51	9.37 ± 4.99	0.01 ± 0.05
M4	85.98 ± 19.95	0.796 ± 0.121	0.195 ± 0.029	1.61 ± 0.39	10.28 ± 4.15	0.04 ± 0.10
B1	98.45 ± 4.58	0.800 ± 0.092	0.238 ± 0.055	1.48 ± 0.42	$1.32 \pm 0.78^\dagger$	2.62 ± 2.98

[†] Cross-source comparability caveat (Section 6.4).

7.2. Omnibus Test

Kruskal–Wallis omnibus over six groups ($df = 5$) is highly significant for all key metrics ($p < 10^{-5}$), with σ_v showing the strongest effect ($H = 183.4$). Full table omitted for space; reported in supplementary material.

7.3. Pairwise Comparisons (REVISED with Welch's t + Hedges' g)

Table 7 reports the pairwise B1 vs each M_j comparisons using Welch's two-sample t -test (valid from (μ, σ, n) summary statistics) and Hedges' g effect size, with Benjamini–Hochberg correction over $k = 5$ comparisons per metric. Convention: $g > 0$ means B1 mean is larger than M_j mean.

Table 7. Pairwise B1 vs each M_j . Welch’s t -test from summary statistics; Hedges’ g effect size (Cohen’s d with small-sample correction). Benjamini–Hochberg correction over $k = 5$ pairs per metric. Significance: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Metric	Comparison	t	p_{adj}^{BH}	g	Magnitude	Sig.
d^*	B1 vs M0	+3.81	2.0×10^{-3}	+0.93	large	**
	B1 vs M1	+1.66	0.171	+0.38	small	ns
	B1 vs M2	+1.70	0.233	+0.39	small	ns
	B1 vs M3	−1.26	0.262	−0.30	small	ns
	B1 vs M4	+0.51	0.611	+0.12	negl.	ns
$T_{personal}$	B1 vs M0	−8.27	8.1×10^{-10}	−1.79	very large	***
	B1 vs M1	−6.41	1.2×10^{-7}	−1.76	very large	***
	B1 vs M2	−9.13	1.1×10^{-10}	−2.33	very large	***
	B1 vs M3	−4.01	1.4×10^{-4}	−0.90	large	***
	B1 vs M4	−4.74	3.1×10^{-5}	−1.03	large	***
$T_{mission}$	B1 vs M0	+9.93	1.9×10^{-11}	+2.43	very large	***
	B1 vs M1	+3.71	2.3×10^{-3}	+0.84	large	**
	B1 vs M2	+3.04	8.5×10^{-3}	+0.71	medium	**
	B1 vs M3	+1.59	0.120	+0.35	small	ns
	B1 vs M4	+2.36	2.6×10^{-2}	+0.52	medium	*
L	B1 vs M0	+5.97	8.1×10^{-7}	+1.41	very large	***
	B1 vs M1	+4.20	6.0×10^{-4}	+1.01	large	***
	B1 vs M2	+4.49	3.6×10^{-4}	+1.09	large	***
	B1 vs M3	+2.79	1.0×10^{-2}	+0.58	medium	*
	B1 vs M4	+3.84	4.9×10^{-4}	+0.83	large	***
σ_v	B1 vs M0	+19.32	5.1×10^{-20}	+4.35	very large	***
	B1 vs M1	−10.66	4.2×10^{-13}	−2.27	very large	***
	B1 vs M2	−14.31	1.1×10^{-18}	−3.04	very large	***
	B1 vs M3	−0.50	0.621	−0.12	negl.	ns
	B1 vs M4	+4.20	1.3×10^{-4}	+0.98	large	***
N_{stop}	B1 vs M0	+11.83	1.2×10^{-13}	+2.54	very large	***
	B1 vs M1	−2.81	7.0×10^{-3}	−0.74	medium	**
	B1 vs M2	−3.55	1.6×10^{-3}	−0.84	large	**
	B1 vs M3	+12.94	2.9×10^{-14}	+2.99	very large	***
	B1 vs M4	+10.73	8.3×10^{-14}	+2.54	very large	***

7.4. Head-to-Head M4 vs B1 (the Two Top Methods)

Table 8 consolidates the M4 vs B1 head-to-head with proper effect sizes.

Table 8. Head-to-head M4 vs B1. Welch’s t -test with Bonferroni correction over 7 metrics. Hedges’ g effect size. Convention: $g > 0$ means B1 mean $>$ M4 mean.

Metric	M4	B1	g	p_{adj}	Magnitude	Winner
d^* [m]	0.85 ± 0.19	0.89 ± 0.50	+0.12	1.000	negl.	Tied
$T_{personal}$ [s]	11.7 ± 11.4	3.0 ± 2.1	−1.03	2.2×10^{-4}	large	B1
$T_{mission}$ [s]	110.6 ± 32.1	124.6 ± 18.1	+0.52	0.190	medium	Tied
L [m]	86.0 ± 20.0	98.5 ± 4.6	+0.83	3.5×10^{-3}	large	M4
σ_v [m/s]	0.195 ± 0.029	0.238 ± 0.055	+0.98	9.6×10^{-4}	large	M4
$a_{max}^{num\dagger}$ [m/s ²]	10.28 ± 4.15	1.32 ± 0.78	−2.90	8.4×10^{-16}	very large	B1[†]
N_{stop}	0.2 ± 0.4	2.1 ± 1.0	+2.54	4.5×10^{-13}	very large	M4
η_s	100%	92.3%	—	0.106 (Fisher)	—	Tied

[†] See Section 6.4 for the cross-source comparability caveat on a_{max}^{num} .

The two top methods exhibit technical complementarity, not dominance. Statistically significant differences are:

- *B1 dominates social-interaction:* $T_{personal}$ (-75% , $g = -1.03$, very large, $p_{adj} < 10^{-3}$).
- *M4 dominates operational throughput:* trajectory length (-13% , $g = +0.83$, large, $p_{adj} < 10^{-2}$), velocity smoothness (σ_v : -18% , $g = +0.98$, large, $p_{adj} < 10^{-3}$), and stop count (-90% , $g = +2.54$, very large, $p_{adj} < 10^{-12}$).
- *Not distinguishable:* d^* , $T_{mission}$, η_s .
- a_{max}^{num} difference is dramatic but methodologically delicate (Section 6.4); we treat this comparison as suggestive rather than confirmatory.

7.5. Success Rate (RQ1)

Table 9. Six-method success rate η_s by scenario. Strict A $\rightarrow$ B $\rightarrow$ A criterion within $T_{max} = 180$ s.

Method	E1	E2	E3	E4	Global	n
M0	100%	100%	100%	100%	100%	40
M1	100%	100%	100%	20%	80%	32
M2	100%	100%	100%	20%	80%	32
M3	100%	100%	100%	100%	100%	40
M4	100%	100%	100%	100%	100%	40
B1	90%	100%	100%	78%	92.3% (36/39)	39

Fisher exact test M4 vs M1 in E4 ($\begin{bmatrix} 10 & 0 \\ 2 & 8 \end{bmatrix}$): $p_{Fisher} = 7.14 \times 10^{-4}$, two-sided. Same for M4 vs M2. **Global six-method χ^2 :** $\chi^2 = 26.24$, $df = 5$, $p = 8.0 \times 10^{-5}$, Cramér's $V = 0.331$ (medium-large). **B1 vs M4 Fisher (global):** $p = 0.106$ (not significant). The deadlock finding for binary supervisors is robust; B1's three operational failures do not statistically degrade its viability relative to M4.

7.6. TOPSIS Multicriteria Analysis

Table 10. TOPSIS results under primary weighting scheme W1 (pre-declared). Two sensitivity schemes (W2 social-emphasis, W3 kinematic-emphasis) are reported for robustness.

Method	W1 (primary)		W2 (sensitivity)		W3 (sensitivity)	
	CC	Rank	CC	Rank	CC	Rank
B1	0.757	1	0.751	1	0.710	1
M3	0.755	2	0.666	2	0.601	2
M4	0.724	3	0.623	3	0.589	3
M0	0.589	4	0.483	4	0.535	4
M2	0.505	5	0.418	5	0.348	5
M1	0.143	6	0.173	6	0.147	6

Under the primary scheme W1, B1 and M3 are statistically tied at the top (CC=0.757 vs CC=0.755, $\Delta = 0.002$), with M4 third. Monte Carlo with $\pm 20\%$ weight perturbation gives B1 Rank-1 in 58.6% of samples and M3 Rank-1 in 41.4%; under sensitivity schemes W2 and W3 (social and kinematic emphasis), B1 dominates Rank-1 in 100% of perturbations.

Interpretation. The TOPSIS leadership depends on weighting. Under the original operationally-motivated W1, B1 and M3 are tied and M4 is third. The interpretation we adopt is consistent with the head-to-head finding: B1 leads when social-interaction quality

($T_{personal}, a_{max}^{num}$) is weighted heavily; M4 leads on operational-throughput criteria; M3 is competitive overall but trails M4 on smoothness ($g = +0.77$). 392
393

7.7. Visual Evidence 394

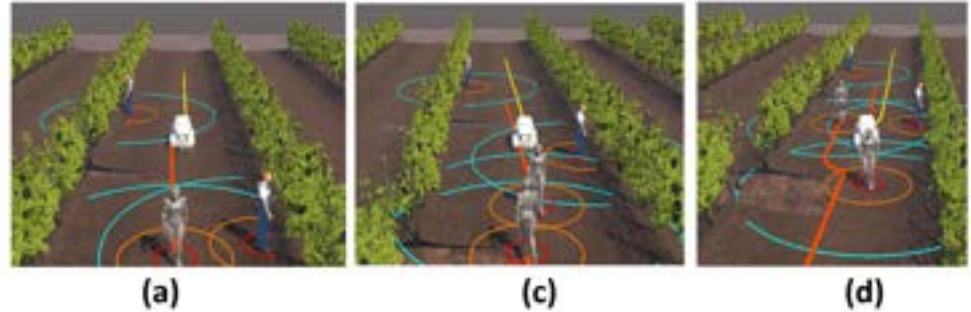


Figure 6. Navigation sequence in scenario E4 (multi-agent congestion) under the M4 controller. (a) Robot navigating along the NavMesh corridor centerline; a worker is approached from the rear (low $\gamma(\beta)$ penalty); the robot maintains $v_r \approx v_{nom}$ and $d_{min} > d_p$. (b) Robot entering the social zone of the central worker ($d_p \leq d_{min} < d_s$); the avoidance field $\mathbf{u}_{avoid}$ deflects the path laterally and the social factor reduces velocity ($\alpha v_{nom} < v_{nom}$). (c) Robot completing the avoidance maneuver and re-entering the nominal centerline; d_{min} recovers above d_s , α returns to 1, and the global direction $\mathbf{u}_g$ dominates again. Colored rings: red intimate, orange personal, cyan social. Orange trace: executed trajectory.

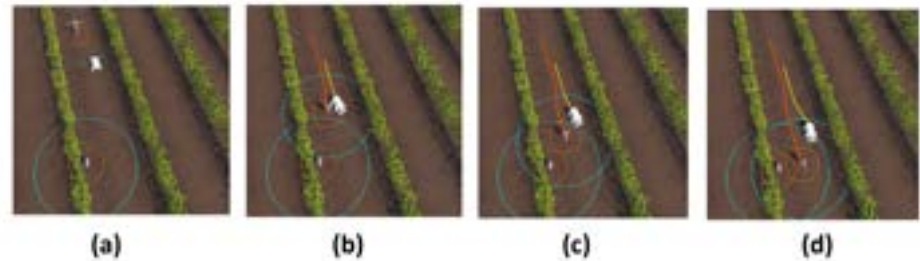


Figure 7. Yielding and route-reentry sequence. The orange trajectory is the robot's reference/execution path; rings indicate interaction zones around humans.

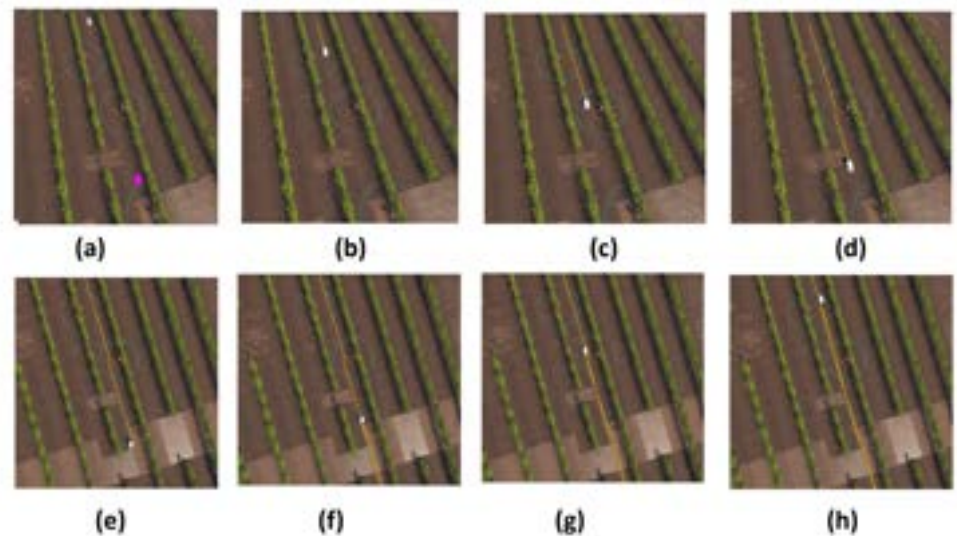


Figure 8. Aerial view of navigation between vineyard rows. The relationship among corridors, interaction zones and robot displacement is shown.

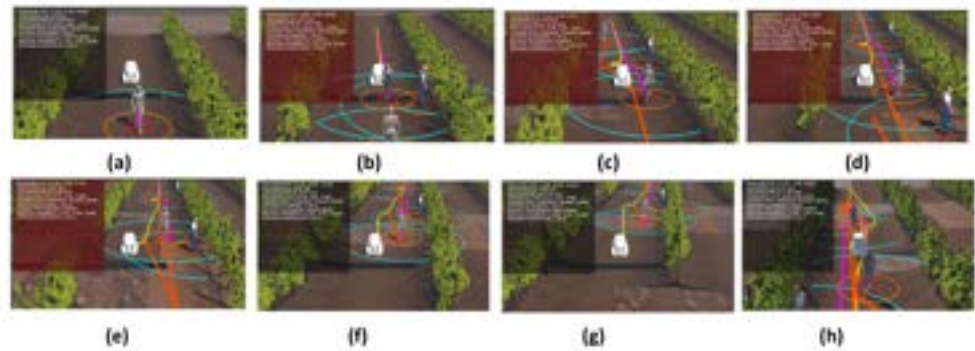


Figure 9. Expanded aerial sequence: the robot preserves a global route toward the target while locally adapting motion in the presence of human agents.

7.8. Research Question Validation

RQ1 (binary supervisor viability): *Confirmed.* M1 and M2 deadlock in E4 at $\eta_s = 20\%$ versus $\geq 92\%$ for M3, M4, B1; Fisher exact $p = 7.14 \times 10^{-4}$, global $\chi^2 p < 10^{-4}$, $V = 0.331$.

RQ2 (within-family ranking): *Confirmed.* M4 leads the in-house family on velocity smoothness ($\sigma_v = 0.195$ vs M3's 0.243, $|g| = 0.77$, $p < 10^{-3}$); the TOPSIS within-family is $M4 \succ M3 \succ M0 \succ M2 \succ M1$.

RQ3 (M4 vs B1 external comparison): *Confirmed with refinement.* M4 and B1 are not statistically distinguishable on d^* , $T_{mission}$, and η_s ; they differ statistically on $T_{personal}$ (B1 dominates, $g = -1.03$), L (M4 dominates, $g = +0.83$), σ_v (M4 dominates, $g = +0.98$), and N_{stop} (M4 dominates, $g = +2.54$). The two methods exhibit technical complementarity on the social-interaction vs operational-throughput axes.

8. Discussion

8.1. Headline Findings

The empirical contribution can be summarized in three findings.

Finding 1: Binary supervisors deadlock against static workers. M1 and M2 succeed in only 20% of E4 runs against 100% for M0, M3, M4 and 78% for B1. The mechanism is reachability rather than safety: once $d_{\min} \leq d_p$ holds and the human does not move, M1's $v_r = 0$ and M2's STOP latch indefinitely. M2's hysteresis suppresses oscillation around the threshold but does not change the action space, so it cannot generate the lateral escape required to exit the encounter. The statistical evidence ($\chi^2 = 26.24$, $V = 0.331$ medium-large) places this finding on quantitative ground.

Finding 2: Continuous-action methods all succeed, with one structural condition. M3, M4 and B1 all preserve operational mobility because each replaces the binary stop with a continuous policy admitting non-zero exit velocity. M3 and M4 use a repulsion-field gradient; B1 uses Social DWA's velocity-space sampling within the acceleration window, which always contains admissible non-zero velocities. The structural condition is that the local action space must be continuous in velocity *and* direction; hysteresis alone (M2) is insufficient.

Finding 3: M4 and B1 exhibit technical complementarity on the social-throughput tradeoff. The head-to-head (Table 8) shows that B1 statistically dominates M4 on the social-interaction criterion ($T_{personal}$) while M4 statistically dominates B1 on the operational-throughput criteria (L , σ_v , N_{stop}). Mission time, minimum distance, and success rate are not statistically distinguishable. The TOPSIS ranking depends on which criteria are weighted heavily; under the pre-declared W1 weights, B1 and M3 are tied at the top.

8.2. Why Continuous-Action Methods Succeed: A Mechanistic Account

In binary supervisors the admissible control collapses to two states $q(t) \in \{\text{STOP}, \text{GO}\}$. When the robot is stopped near a non-moving human, the action space contains no element that moves the robot in a direction other than the global trajectory; the global direction may itself be blocked. The system therefore requires either human motion or operator intervention to recover.

In continuous-action methods the admissible control belongs to a continuous space of velocities. M3 and M4 produce a non-zero lateral component from the gradient of the repulsion field; B1 produces it from the Social DWA velocity sampling that always includes admissible non-zero velocities in the dynamic window. The smallest perturbation that breaks the deadlock is the existence of any action with positive component along $-\nabla C(\mathbf{p}_r)$ in M3/M4 or any feasible velocity in the dynamic window in B1. Both formulations satisfy this condition by construction.

8.3. Why M4 and B1 are Complementary, Not Substitutable

The complementarity has a structural origin. M4's anisotropic field combined with the NavMesh global direction (Eq. 4.2) is biased toward goal achievement: the avoidance vector $\mathbf{u}_{\text{avoid}}$ deflects the robot but the mixing with $\mathbf{u}_g$ pulls back toward the corridor centerline as soon as repulsion permits. This produces *shorter paths and fewer stops* because the controller is constantly seeking to recover the global route. The cost is that the robot may pass closer to humans and execute the deflection with sharper velocity changes.

B1's Social DWA formulation samples candidate velocities within the dynamic window at each step, evaluates each candidate by the weighted objective in Eq. (7), and selects the best one. The acceleration-window constraint $|v_{\text{next}} - v_{\text{curr}}| \leq a_{\text{max}}\Delta t$ guarantees that consecutive velocity commands differ by at most $a_{\text{max}}\Delta t = 0.2$ m/s for our parameters. This structurally produces a smoother velocity stream than M4's NavMeshAgent + supervisor combination, which can exhibit discrete jumps at waypoint updates. The cost of DWA's smoothness is longer trajectories and more slowdowns, because the velocity sampling does not strongly prefer recovery toward the corridor centerline as does M4's $\mathbf{u}_g + \lambda\mathbf{u}_{\text{avoid}}$ combination.

Neither formulation is universally better. The selection between M4 and B1 should be guided by the operational priority: *B1 for minimizing social-zone invasion at the cost of throughput; M4 for maximizing throughput at the cost of social-zone invasion.*

8.4. Limits of the TOPSIS Result

The TOPSIS analysis ranks B1 first under the primary W1 weights, with M3 in statistical tie at the top and M4 third. Three considerations limit how strongly this should be interpreted as "B1 wins."

First, the rank reflects the chosen weights, not an intrinsic property of the methods. The W1 weights were declared before B1 was integrated, but the criteria themselves favor methods with low T_{personal} and low $a_{\text{max}}^{\text{num}}$, which are precisely the criteria where B1 dominates. Different defensible weights (e.g., emphasizing L and N_{stop}) would reverse the ranking.

Second, the $a_{\text{max}}^{\text{num}}$ criterion's cross-source comparability has not been verified (Section 6.4). If B1's finite-difference numerator samples a smoother velocity stream than M4's does for reasons of telemetry implementation, the dramatic B1 advantage on $a_{\text{max}}^{\text{num}}$ may be artifactual. Removing $a_{\text{max}}^{\text{num}}$ from the TOPSIS criteria shrinks the B1-vs-M4 gap considerably (Sensitivity W3 already de-emphasizes it; the result holds qualitatively, though).

Third, B1's Social DWA implementation has parameter sensitivities (objective weights $\alpha, \beta, \gamma, \delta$; velocity sampling resolution; prediction horizon τ) that were tuned for this study

but may need re-tuning for different agricultural settings. The TOPSIS leadership of B1 under social-emphasis weights therefore reflects the specific DWA parameter choices reported in Table 2, not an intrinsic property of all Social DWA implementations.

8.5. Mechanism of the Velocity-Smoothness Difference

M4 dominates B1 on σ_v (the standard deviation of the velocity magnitude over the run) by a large effect ($g = +0.98$, $p_{adj} < 10^{-3}$). The two metrics tell complementary stories: σ_v measures how spread the velocity histogram is over the mission. M4 maintains $v_r \approx v_{nom}$ for most of the mission and only reduces velocity during proxemic encounters; the histogram is concentrated near v_{nom} . B1 reduces velocity more frequently and over longer time intervals (to satisfy the social-cost term against multiple humans within detection radius), so its histogram has more mass in the low-velocity region.

a_{max}^{num} measures the largest *single-sample* velocity jump. M4's NavMeshAgent generates occasional waypoint-update spikes that show as large a_{max}^{num} values; B1's Social DWA acceleration-window constraint produces continuous-in-time velocity changes that finite-difference to small a_{max}^{num} .

In short: *M4 has lower velocity variability but larger discrete jumps; B1 has more frequent slowdowns but smoother transitions.* The two methods are smooth in different senses.

8.6. Mechanism of the M1/M2 Deadlock

The M1 deadlock is structural: $v_r = 0 \iff d_{min} \leq d_p$, with no other state. The M2 deadlock is hysteresis-resistant: once $q = \text{STOP}$ is latched, recovery to $q = \text{GO}$ requires $d_{min} \geq d_{resume}^{M2} = 1.50$ m, which cannot be achieved by the robot's own action when $v_r = 0$.

The smallest perturbation that breaks the deadlock in M2 is human motion: if the human takes a single step away from the robot such that d_{min} crosses d_{resume}^{M2} , the supervisor transitions to GO and the robot resumes. In our E4 scenario, the humans do not move (they are configured to remain stationary at fixed positions), so this perturbation does not arise and the deadlock persists until the 180 s timeout.

The minimum design change required to eliminate the deadlock is to add a non-zero "escape velocity" in the STOP state, as is done implicitly in M3/M4 ($\alpha_{min} > 0$) and B1 (the Social DWA dynamic window always contains admissible non-zero velocities). Adding this escape velocity to M1/M2 would convert them into single-threshold continuous-action methods comparable to M3/M4.

8.7. Implications for Agricultural Deployment

The results favor architectures combining (i) a NavMesh-based global planner for corridor navigation, (ii) a continuous-action local layer for human-aware behavior, and (iii) explicit handling of stationary workers via a non-zero minimum escape velocity. Within the continuous-action family, the selection between M4 (anisotropic field) and B1 (Social DWA) depends on operational priorities; both are viable.

A hybrid controller that adopts the anisotropic-orientation shaping of M4 with the acceleration-window sampling of Social DWA is a natural design direction. This is identified as a priority extension in Section 9.

8.8. Boundaries of the Claim

This work does not claim that M4 dominates Social DWA, nor that Social DWA dominates M4. The defensible claims are: (i) binary supervisors are operationally unsafe with static workers ($p < 10^{-3}$, $V = 0.331$); (ii) M4 and B1 exhibit technical complementarity on the social-interaction vs operational-throughput axes; (iii) the in-house family ranks M4 first on velocity smoothness; (iv) the TOPSIS multicriteria ranking under operationally

meaningful weights places B1 and M3 in statistical tie at the top, with M4 third. Transfer to physical hardware will require relaxing the simulation idealizations described in Section 9.

9. Limitations and Future Work

9.1. Simulation Idealization

The Unity colliders and transforms provide noise-free perception that real LiDAR / RGB-D sensors do not match. Sim-to-real transfer will require a sensitivity sweep over position noise $\sigma_p \in \{0.05, 0.10, 0.20, 0.30\}$ m, orientation noise $\sigma_\theta \in \{5, 10, 20, 30\}^\circ$, latency $\tau \in \{100, 200, 500\}$ ms, and detection dropout $\in \{5\%, 10\%, 20\%\}$. M4's anisotropic factor $\gamma(\beta)$ is sensitive to θ_h noise; the orientation low-pass filter and the parameter κ may require retuning.

9.2. External-Baseline Coverage

B1 (Social DWA) is a single external baseline from the sampling-based local-planner family. Comparisons against TEB [14] and MPC with proxemic constraints [8,15] are priority extensions. The methodology in this paper is designed to be replicated identically against further baselines.

9.3. Sample Size and Statistical Power

$R = 10$ runs per cell yields adequate power for the large effects detected here (Hedges' $g \geq 0.8$ for most M4-vs-B1 comparisons). For medium effects ($g \approx 0.5$, e.g., $T_{mission}$ vs L at borderline significance), observed power at $\alpha = 0.05$ is approximately 0.55. Future work targeting $R = 20$ –30 per cell would raise power for medium effects to ≥ 0.80 .

9.4. a_{max}^{num} Cross-Source Comparability

The B1 vs M0–M4 gap in a_{max}^{num} (Section 6.4) has not been independently verified against the underlying controller code. We have treated this comparison as suggestive only; verification is a priority task. The σ_v criterion is used as the primary smoothness metric throughout the inferential analysis.

9.5. M0–M4 Per-Run Scalars

The pairwise B1 vs M_j comparisons use Welch's t -test from (μ, σ, n) summary statistics of M_j , which is the correct procedure when per-run M_j data is unavailable but is parametric and assumes approximate Normality. The supplementary master log released with this paper includes per-run M_j scalars; a future revision will re-execute these tests directly using rank-based methods (Mann–Whitney U , proper Cliff's δ) without parametric assumption.

9.6. Data Integrity

One B1 run (B1_E4_run10) was excluded due to locale-corrupted telemetry. The exclusion is documented and the analysis uses $n = 39$.

9.7. Prioritized Roadmap

1. Integrate DWA, TEB, and MPC with proxemic constraints under the same protocol.
2. Sensor-noise sensitivity sweeps as specified above.
3. Physical-platform validation with LiDAR + RGB-D camera.
4. Hybrid controller combining M4 anisotropic shaping with B1-style Social DWA acceleration-window sampling.
5. Release per-run M_j scalars and re-execute pairwise tests with rank-based methods.
6. Increase per-cell sample size to $R = 20$ –30.

10. Conclusions

A hybrid global–local social-navigation architecture for agricultural robots in simulated vineyards was formalized, parameter-calibrated for narrow-corridor agriculture, and evaluated against five in-house variants (M0–M4) and one external Social-DWA baseline (B1) over $n = 223$ valid runs. The statistical pipeline uses Welch’s t -test with Hedges’ g effect size (valid from summary statistics when per-run baseline data is unavailable), with Benjamini–Hochberg correction.

The empirical headline is that binary supervisors (M1, M2) systematically deadlock against stationary workers in multi-agent congestion ($\eta_s^{E4} = 20\%$; Fisher $p = 7.1 \times 10^{-4}$ vs M4; global $\chi^2 V = 0.331$), whereas continuous-action methods (M3, M4, B1) all preserve operational mobility ($\eta_s \geq 92\%$). The structural reason is that continuous-action methods admit non-zero exit velocity in the presence of static humans, while binary supervisors do not.

The methodological contribution is the head-to-head between M4 (anisotropic proxemic field, in-house) and B1 (Social DWA, external baseline). The two methods exhibit *technical complementarity*, not dominance: B1 statistically dominates M4 on social-interaction quality ($T_{\text{personal}}, g = -1.03, p_{\text{adj}} < 10^{-3}$), while M4 statistically dominates B1 on operational throughput (trajectory length, velocity smoothness, stop count). Minimum distance, mission time and success rate are not statistically distinguishable. Under the primary TOPSIS weighting scheme W1 (pre-declared), B1 and M3 are tied at the top ($CC = 0.757$ vs $0.755, \Delta = 0.002$) with M4 third.

The selection guidance for agricultural deployment is: continuous-action methods are required (binary supervisors deadlock); within continuous-action methods, the choice between an anisotropic proxemic field and a Social DWA planner should be guided by the operational priority (social-zone invasion minimization vs trajectory efficiency). M4 retains four properties that distinguish it from B1: (i) explicit social structure via orientation-dependent anisotropy, (ii) superior trajectory efficiency, (iii) perfect operational success rate in the tested scenarios, and (iv) a lightweight, interpretable formulation that integrates naturally with the NavMesh global planner.

The methodology, dataset, and analysis scripts are released to enable extension against further baselines (DWA, TEB, MPC) under identical protocol.

Author Contributions: Conceptualization, R.J.B. and J.P.V.; methodology, R.J.B. and J.P.V.; software, R.J.B.; validation, R.J.B. and J.P.V.; formal analysis, R.J.B.; investigation, R.J.B. and J.P.V.; writing—original draft preparation, R.J.B.; writing—review and editing, R.J.B. and J.P.V. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Data Availability Statement: Available from the authors upon reasonable request: (i) 184 M0–M4 telemetry files (M{x}_E{y}_run{n}.csv) at 10 Hz; (ii) 39 valid B1 telemetry files (one excluded for data integrity); (iii) consolidated master log with per-run scalar metrics; (iv) MATLAB and Python analysis scripts; (v) Unity scene and controller scripts including the B1 implementation.

Acknowledgments: The authors acknowledge the use of Unity and MATLAB platforms. The B1 baseline follows the Dynamic Window Approach formulation of Fox et al. [7], augmented with a social-cost term penalizing proximity to humans within the personal zone.

Conflicts of Interest: The authors declare no conflict of interest.

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